

BDAI PROJECT OVERVIEW

Leveraging Bangladesh Sectoral Knowledge Graphs & LLMs for AI-Driven Insights

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Mission Statement: To leverage **Knowledge Graphs (KGs)** and **Large Language Models (LLMs)** to develop AI-powered natural language interfaces. This ecosystem enables powerful, data-driven insights across Bangladesh's **socio-economics, agriculture, healthcare, tourism, and education** sectors, directly empowering informed decision-making and accelerating progress toward the **Sustainable Development Goals (SDGs 2, 4, 9, 10, 17)**.

Core Objectives & Innovations

- **SMART Data Ecosystem:** A robust architecture ensuring data is *Sustainable, Meaningful, Accessible, Reusable, and Traceable*.
- **Sector-Specific KGs:** Construction and optimization of domain-specific graphs (Socio-Economic, Agriculture, Healthcare, Education, Tourism).
- **Open Data Framework:** Establishing a comprehensive, localized open data quality measurement framework.
- **askBDAI System:** An advanced AI-powered Q/A interface exploiting semantic graph patterns for intuitive data exploration.
- **KG-RAG Architecture:** Enhancing LLM precision by leveraging Knowledge Graphs for context in Retrieval-Augmented Generation pipelines.
- **Explainable AI:** Utilizing KGs to validate LLM outputs, ensuring fairness, reducing bias, and maintaining traceable logic.

Underlying Technologies

- **Semantic Web Layer:** Intensive utilization of **RDF**, **Ontologies**, and **Linked Open Data (LOD)** principles for robust data modeling.
- **Data Federation:** Ensuring **FAIR** (Findable, Accessible, Interoperable, Reusable) compliance across diverse multi-sector datasets.
- **Advanced Analytics:** Supporting descriptive, diagnostic, predictive, and prescriptive analytics.

System Architecture

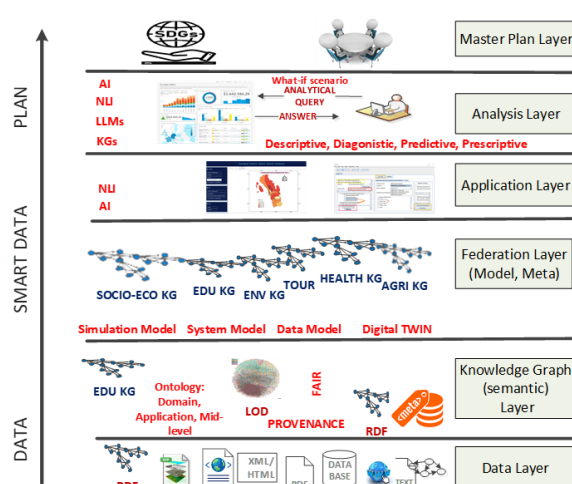


Fig: The multi-layered BDAI architecture from Data Layer to Master Plan Layer.

Target Beneficiaries

- **Researchers & Students:** Direct engagement with cutting-edge Semantic Web tools, HPC infrastructure, and LLM pipelines.
- **Government:** Access to actionable, data-driven policy insights.
- **General Public:** Tangible, AI-driven solutions addressing real-life socio-economic and public service challenges.